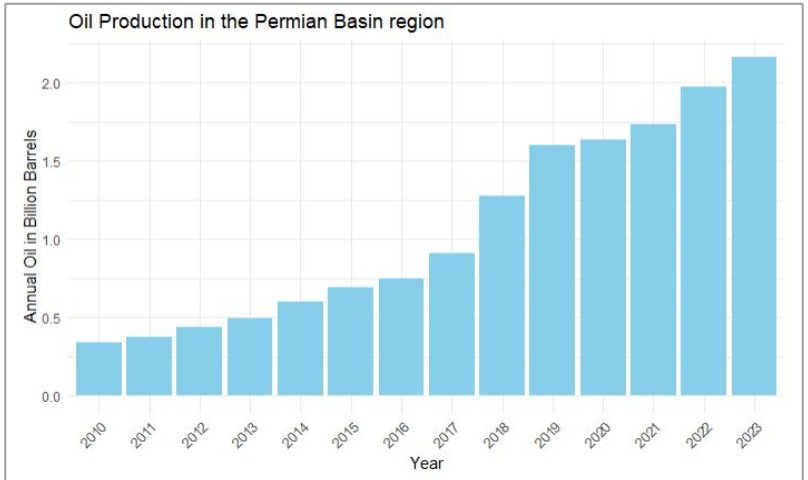
Estimating the association between multi-frequency noise measurements and sources of air pollution from oil and gas operations

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Background

- The Permian Basin, straddles Texas and New Mexico, is one of the most profitable and productive fracking locations in the world.

- Annual oil production has increased over 8-fold since 2010 at wells located in the Permian Basin.



Background

- Fracking involves injecting high-pressure fluid into shale formations to allow oil and natural gas to flow more freely to the production well.

- However, there are many environmental concerns around fracking
  - noise
  - air pollution
  - greenhouse gas emissions (methane and carbon dioxide)
  - radioactivity

- Goal: find noise frequencies associated with pollutants
  - For flaring source apportionment

Data

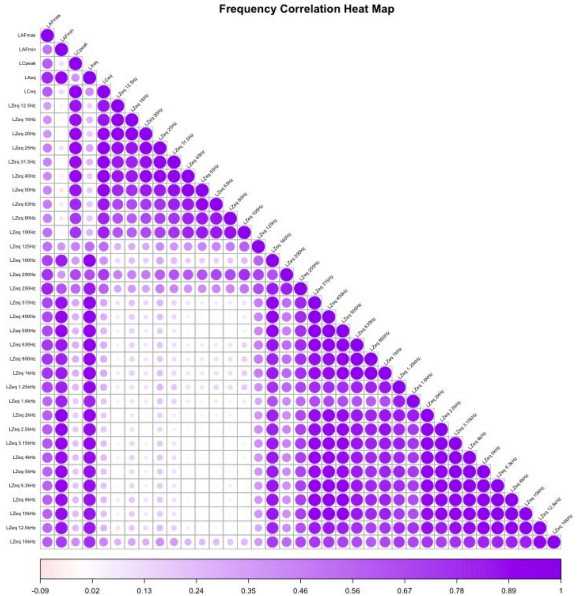
- 1-mins resolution data collected by a stationary monitoring platform, the “trailer”
  - sampling from May 1, 2023 to May 31, 2024
  - 30 noise frequencies from 12.5Hz to 16kHz, with 5 weighted frequencies
  - pollutants & greenhouse gasses: Nitrogen Oxides (NOx), Methane, etc





Methodology

Noise Frequency I

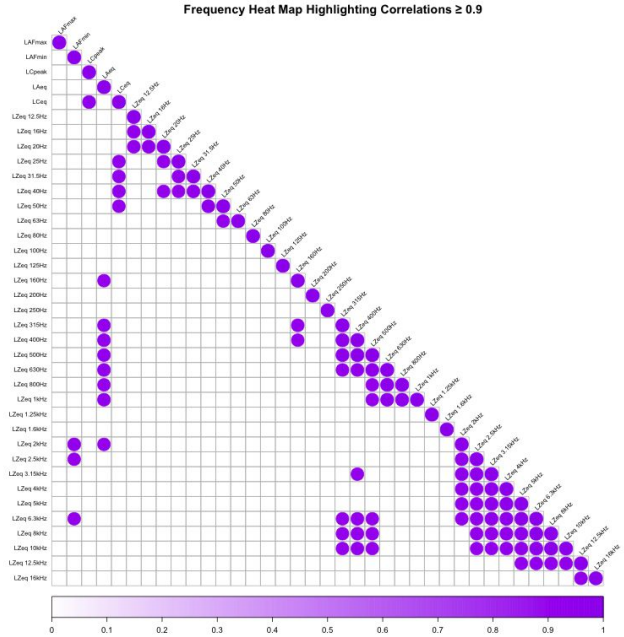


Correlation heatmap of all 30 frequencies and the 5 weighted measures

Correlation heatmap shows groupings of low and high frequencies.

→ highlight and groups where frequencies correlate > 0.9.

→ create differenced bands to help identify sources.



Correlations > 0.9

Noise Frequency II

Grey, blue, purple, and single frequencies are lower-frequency potentially representing compressors, generators, drilling. Sounds can travel long distances and can cause vibrations.

Weighted measures:

- LAeq reflects human perception, weights in mid frequencies more.
- LAFmax, LAFmin represent the max and min A-weighted fast response time (capture sub-minute sound pressure changes).
- LC reflects weights low frequencies more, useful for engines, industrial operations.
- LCpeak measures loudest peak in low frequency.

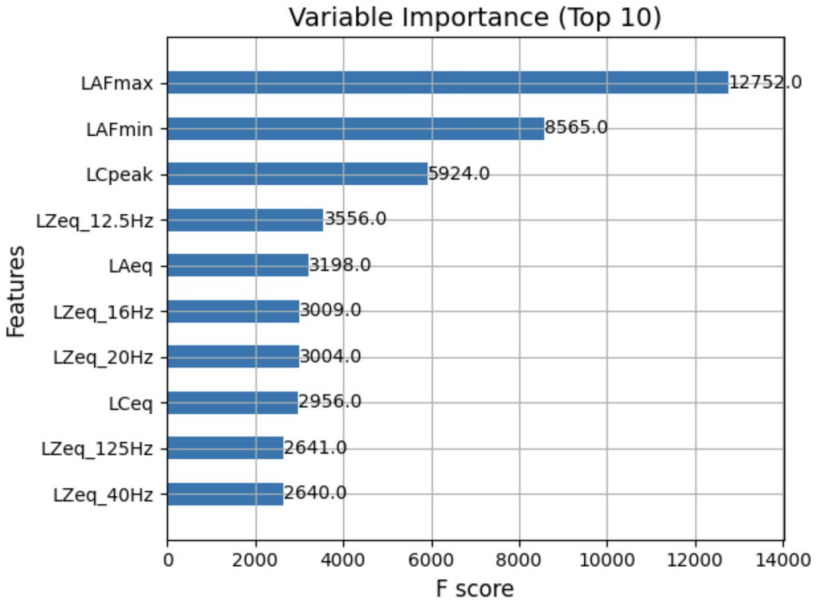
Frequency	Frequency	Noise indicator	Difference Frequency	Metrological variable (station)
LZeq 12.5Hz	LZeq 500Hz	LAFmax	LCpeak - LCeq	Ambient Temp (F)
LZeq 16Hz	LZeq 630Hz	LAFmin	LAFmax - LAFmin	Wind Speed (m/s)
LZeq 20Hz	LZeq 800Hz	LCpeak	LZeq 25Hz -500Hz	Wind Direction (degrees)
LZeq 25Hz	LZeq 1kHz	LCeq	LZeq 250Hz -800Hz	Relative Humidity (%)
LZeq 31.5Hz	LZeq 1.25kHz	LAeq	LZeq 250Hz -500Hz	Pressure (Pa)
LZeq 40Hz	LZeq 1.6kHz		LZeq 500Hz -800Hz	Rainfall (mm)
LZeq 50Hz	LZeq 2kHz		LZeq 500Hz -5KHz	Solar Radiation (W/m²)
LZeq 63Hz	LZeq 2.5kHz		LZeq 500Hz -12.5KHz	
LZeq 80Hz	LZeq 3.15kHz			Down/Up wind (derived)
LZeq 100Hz	LZeq 4kHz			Day/Night (derived)
LZeq 125Hz	LZeq 5kHz			
LZeq 160Hz	LZeq 6.3kHz			
LZeq 200Hz	LZeq 8kHz			
LZeq 250Hz	LZeq 10kHz			
LZeq 315Hz	LZeq 12.5kHz			
LZeq 400Hz	LZeq 16kHz			

Yellow and single frequencies are mid-frequency potentially representing pumps, machinery, pipelines. Sounds are audible.

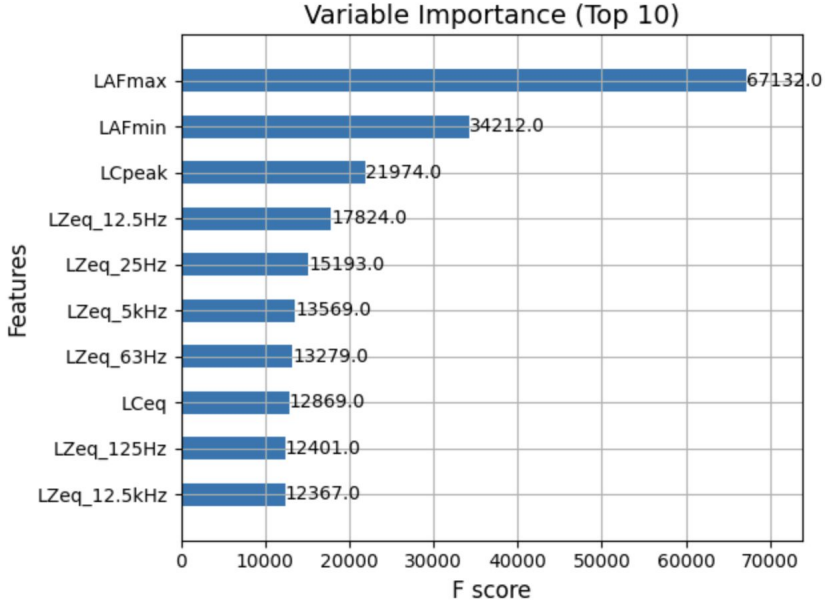
Red and grey frequencies are high-frequency signatures potentially representing flaring and venting. Sounds are audible and disturbing.

NOx Prediction: 2 XGBoost Models

1. Full Frequency Range
   - R-squared: 0.43



2. Selected Frequencies from grouping
   - R-Squared: 0.67



Conclusion?

Future Plan

- filter out the effect of air conditioner turning on and off
  - potential contamination in low to mid-frequency range



Thanks! Any Questions?!